Seungmin Lee

Department of Mathematics, Florida State University
Applied and Computational Mathematics | sl22b@fsu.edu

Education

Ph.D., Applied and Computational Mathematics August 2022 - present
Florida State University, Tallahassee, Florida, USA

M.S., Mathematics March 2020 - February 2022
Dongguk University, Seoul, South Korea
Thesis: Neural Network in Solving Ordinary Differential Equations

B.S., Mathematics March 2016 - February 2020
Dongguk University, Seoul, South Korea

Research Experience

Research Assistant, Florida State University Fall 2026; Spring 2025

Intern, Lawrence Livermore National Laboratory (LLNL)
June 3 - July 13, 2024

Publications and Manuscripts

Published

[1] **Seungmin Lee** and Sanghyun Lee.
Convergence and numerical simulations of the continuous data assimilation for Biot's poroelasticity system.
Computers & Mathematics with Applications, 202:20-37, 2026.
doi:10.1016/j.camwa.2025.11.008

Submitted

[2] Sanghyun Lee, **Seungmin Lee**, and Chen Liu.
An optimization-based property-preserving limiter for two-phase flow in porous media on non-matching meshes.
Submitted. arXiv:2610.33940

[3] **Seungmin Lee**, Sanghyun Lee, and Chen Liu.
Optimization-based cell average limiters as metric projections: Constrained evolution and convergence analysis for convection-diffusion equations.
Submitted. arXiv:2610.33449

Ongoing Work

[4] Sanghyun Lee, **Seungmin Lee**, Baehyun Min, and Joshua White.
A Bound-Preserving Discretization and Block Preconditioning for Compressible Two-Phase Flow in Deformable Porous Media with Application to Underground Hydrogen Storage.

[5] Phuoc Toan Huynh, **Seungmin Lee**, Feng Bao, and Sanghyun Lee.
Ensemble Score Filtering for State Estimation in a Two-Phase Flow Model with Rare Earth Materials.

Teaching Experience

Florida State University

Course Instructor

Calculus II	Spring 2026
Calculus I	Fall 2025

Teaching Assistant

Course	Role	Term(s)
Discrete Mathematics II	Recitation TA	Fall 2024
Calculus I	Recitation TA	Spring 2024
College Algebra	Lab TA	Fall 2022, Spring 2023, Fall 2023
Liberal Arts Mathematics	Lab TA	Fall 2023
Analytic Trigonometry	Lecture TA	Summer 2023
Business Calculus	Lab TA	Fall 2022, Spring 2023

Dongguk University

Grading Assistant

Modern Algebra	March - June 2021
Cryptography	September - December 2020
Calculus I and II	March 2018 - December 2019

Conferences, Workshops, and Seminars

- **Applied and Computational Math Seminar**, Fall 2026.
  Florida State University, Tallahassee, Florida, USA. September 8, 2026.
- **2026 Southeast Applied and Computational Math Student Workshop**.
  Emory University, Atlanta, Georgia, USA. April 3-4, 2026.
- **M3L Team All-Hands Meeting**.
  The University of Texas at El Paso, El Paso, Texas, USA. May 15-17, 2025.
- **2025 Southeast Applied and Computational Math Student Workshop**.
  Auburn University, Auburn, Alabama, USA. April 4-6, 2025.
- **11th deal.II Users and Developers Workshop**.
  Colorado State University, Fort Collins, Colorado, USA. August 12-16, 2024.